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HERITAGE LANGUAGE JOURNAL 18 (2021) 1–32

Heritage Language Journal
brill.com/hlj

# Perception of Portuguese Mid Vowels by Heritage Speakers of Brazilian Portuguese

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## Abstract

The main goal of this study is to investigate the perception of Portuguese mid-vowel contrasts (/ɛ/-/e/ and /ɔ/-/o/) by heritage speakers (HSs) of Brazilian Portuguese (BP) who grew up in the United States. This study shows, for instance, an asymmetry in categorization as a function of degree of language dominance for mid-back vowel contrast, but not for mid-front vowel contrast.

## Keywords

heritage language – perception – vowels – Brazilian Portuguese

## 1 Introduction

It is common to see heritage learners of Portuguese in Portuguese as a foreign language classes in the United States. In fact, heritage speakers (HSs) of Portuguese, along with Spanish speakers, constitute the largest group of students enrolled in Portuguese classes in the United States, 45% and 18%, respectively, of the total number of students (Milleret, 2012). The investigation of the heritage Portuguese language system has important implications, not only for the understanding of heritage language (HL) acquisition, but also for educational and pedagogical practices. The study of phonetics/phonology/pronunciation of Portuguese as an HL is still an open area of investigation with few studies in this area (Flores and Rato (2016) is one of these studies). In fact, phonetics/phonology is still considered an under-researched area in HLs overall as compared to other areas of the linguistic system (e.g., Polinsky & Kagan, 2007). A possible

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explanation for this might be related to a common description of heritage speakers "as having the phonological skills of native speakers, especially when they are compared to adult L2 learners of similar morphosyntactic proficiency" (Montrul, 2016, p. 82). Nevertheless, HSs are also perceived anecdotally as having a slight 'accent' when compared to 'real' speakers of the language (Polinsky & Kagan, 2007). There is some evidence that the heritage phonology system differs from monolingual native speakers (e.g., see Rao and Ronquest, 2015, for a review of studies on the phonology of HSs of Spanish). Nevertheless, the extent to which HL phonology and the baseline overlaps and/or differentiates is still not clear and more studies on the HL phonology are necessary.

Heritage speakers are typically "unbalanced bilinguals" (Polinsky, 2018, p. 4); that is, their HL (i.e., their first language) is the weaker language, whereas their second language is the dominant language. Heritage speakers are therefore native speakers of their HL. HSs experience a language shift when they start attending school; that is, the majority language, learned as a second language (L2), becomes the stronger language, and the HL becomes the weaker language (Montrul, 2016). In addition, some HSs become literate in the HL, while others do not. As a result, HSs' proficiency levels in their HL may vary widely, going from having "merely receptive ability to very fluent speech" (Montrul, 2016, p. 41). Heritage languages are demonstrated to have their own route of development in many ways, which may be related to the context in which the language is acquired. For instance, while HSs acquire the language in naturalistic settings in their childhood, L2 learners generally acquire the L2 later in life and in a more formal setting, such as at school (e.g., Castro, Rothman, & Westergaard, 2016).

Studies on the acquisition of mid-vowel contrasts by HSs of Romance languages other than Portuguese, but closely related (e.g., Catalan and Galician), may provide an overall picture of the dynamics of these contrasts. Such studies have shown that these contrasts pose some challenges for HSs. Whereas a number of studies have demonstrated that the perception and production of mid-vowel contrasts may be affected by language dominance, even when HSs are highly fluent bilinguals (e.g., Amengual, 2015, 2016a, 2016b; Amengual & Chamorro, 2015; Mora & Nadeu, 2012; Sebastián-Gallés and Soto-Faraco, 1999; Simonet, 2014), other studies have shown that this effect may be partial or non-existent (e.g., Mayr, López-Bueno, Fernández, & Lourido, 2019). Studies of Catalan mid-vowel contrasts have focused on early and highly fluent Catalan-Spanish bilinguals (Spanish mid vowels encompass only /e/ and /o/ mid vowels) who are from Majorca (e.g., Amengual, 2015, 2016a, 2016b; Simonet, 2014) and from Barcelona (e.g., Mora & Nadeu, 2012; Sebastián-Gallés & Soto-Faraco, 1999). Catalan speakers from Majorca produce a great acoustic

distance between the mid vowels as compared to those from Barcelona (Central Catalan) (e.g., Recasens & Espinosa, 2006, 2009). Regarding the investigation of Galician mid vowels, Spanish-dominant speakers have shown to produce a single front mid vowel (i.e., a merged vowel) in the production of the Galician mid-vowel contrast (Amengual & Chamorro, 2015), whereas other studies have found no differences in the production of Galician mid-vowel contrasts by Galician-Spanish bilinguals (Mayr et al., 2019). Overall, these studies show that early exposure to the HL may not guarantee a well-defined phonemic categorization of mid vowels by HSs. The present study aims to add to this body of research by investigating these contrasts among HSs of Brazilian Portuguese (BP).

## 2 Portuguese and English Mid Vowels

The Portuguese vowel system distinguishes its mid vowels, produced in stressed position, through two articulatory and acoustic dimensions: height and frontness/backness. There are two high-mid vowels, /e/ (front vowel) and /o/ (back vowel), and two low-mid vowels, /ɛ/ (front vowel) and /ɔ/ (back vowel). These vowels produce minimal pairs (e.g., s/e/de 'thirst' – s/ɛ/de 'headquarters'; av/o/ 'grandfather' – av/ɔ/ 'grandmother'). Cross-dialectal acoustic studies have shown that there are some differences in the production of Portuguese mid vowels in BP and European Portuguese (EP) varieties. Escudero, Boersma, Rauber, and Bion (2009) found, for instance, that in terms of F1, /ɛ/ and /e/ are closer in the EP vowel space than in that of BP, which may indicate that /ɛ/ and /e/ in EP might be going through an "impending merger" (p. 1390). Their BP mid-vowel values are reported on Table 1 for reference.

TABLE 1 Average values (ms and Hz) for BP mid vowels (duration, F0, F1, F2, and F3), produced by male ($n$ = 10) and female ($n$ = 10), reported in Escudero et al. (2009, p. 1383)

| Vowel | Duration | | F0 | | F1 | | F2 | | F3 | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Female | Male | Female | Male | Female | Male | Female | Male | Female | Male |
| /ɛ/ | 141 | 123 | 210 | 124 | 646 | 518 | 2271 | 1831 | 2897 | 2572 |
| /e/ | 122 | 109 | 219 | 131 | 425 | 357 | 2468 | 2028 | 3074 | 2719 |
| /ɔ/ | 139 | 123 | 211 | 122 | 681 | 532 | 1054 | 927 | 2653 | 2335 |
| /o/ | 123 | 110 | 224 | 132 | 442 | 372 | 893 | 804 | 2627 | 2380 |

There is evidence that these contrasts are the last ones to be acquired by native speakers in terms of speech production (Matzenauer & Costa, 2017). Mid-vowel contrasts also have a low functional load; that is, there are relatively few minimal pairs with these contrasts in BP (Alves, 2008). In addition, mid vowels are less frequent in the phonological inventory of languages around the world; for example, mid vowels are affected by asymmetry (the lack of either low-mid front or low-mid back vowels) in different languages (Maddieson, 1984). Mid-vowel contrasts are, therefore, more marked (i.e., less universal, less frequent, more complex).

In English, the vowels /e, ɛ, ɔ, o/ are part of its inventory (e.g., /bejt/ "bait" – /bɛt/ "bet"; /bowt/ "boat" – /bɔt/ "bought"); however, they are not realized in English the same way they are in Portuguese. In English, /e/ and /o/ are typically diphthongized (Nearey, 2013). There is also dialectal variability; for example, Upper Midwest speakers tend to monophthongize /ow/ as [o] (Rose, 2006), and Midwestern and California speakers do not distinguish between /ɑ/ and /ɔ/ (as in "cot" and "caught") (Ladefoged & Johnson, 2011).

The questions to be answered on the acquisition of Portuguese mid vowels by HSs whose are English dominant:

(1) Would differences in levels of experience with the HL affect mid-vowel categorization?;

(2) Would HSs show bias toward one of the mid vowel categorization?

The present study aims to examine some of these issues by investigating the perception of Portuguese /ɛ/-/e/ and /ɔ/-/o/ vowel contrasts by adult HSs of BP who live in the United States.

## 3 The Present Study

The main goal of this study is to investigate the perception of Portuguese mid-vowel contrasts (/ɛ/-/e/ and /ɔ/-/o/) by HSs of BP who were born and/or grew up in the United States. Participants took part in two experiments used to elicit perception data. These perception experiments inform us, for instance, whether participants show categorical identification of Portuguese mid vowels, and whether the perception of one vowel contrast can be more (or less) robust than the other. On a more exploratory side, This study is also interested in investigating whether HSs' language behavior, such as the use of Portuguese and connections with the language and culture, may correlate with their proficiency in Portuguese and language dominance. As a broader goal, this study aims to contribute to the understanding of the phonetic/phonological systems of HLs, an area with fewer studies when compared to other linguistic fields,

such as morphology and syntax (Montrul, 2016). This study can be considered the first step in understanding the perception of mid vowels focusing on this population.

Two research questions guide the present study:

RQ1: Do HSs of BP categorize Portuguese mid-vowel contrasts in perception?

RQ2: Does degree of language dominance affect the perceptual categorization of Portuguese mid-vowel contrasts by HSs of BP?

It is hypothesized that HSs of BP will demonstrate some degree of difficulty in categorizing the contrasts as opposed to homeland speakers. Heritage speakers receive limited input from their families and heritage communities; the language to which HSs are exposed is less homogenous than that of the homeland (Polinsky & Scontras, 2019). As a result, the vowel systems of the HL and the language of the homeland may have systematic differences (e.g., Rao & Ronquest, 2015). Consequently, differences in the categorization of Portuguese mid vowels by HSs and homeland speakers may occur.

In addition, it is also hypothesized that the degree of language dominance will affect how HSs of BP perceive Portuguese mid-vowel contrasts. In the present study, HSs of BP are expected to be mostly unbalanced Portuguese-English bilinguals, since balanced bilinguals among adults who have received frequent input from both languages are much less frequent (Montrul, 2018). As a consequence, HSs' degree of dominance in English may vary. Gertken, Amengual, and Birdsong (2014) argue that the conceptualization of language dominance should be understood in terms of gradation and degrees, not in terms of binary language dominance. In fact, Amengual (2015) showed that the perception of Catalan mid vowels was related to the degree of Spanish dominance; for instance, moderately to strongly Spanish dominant bilinguals had more difficulty in distinguishing mispronounced words from correctly pronounced Catalan words with mid-vowel contrasts. Therefore, in this study, it is hypothesized that BP bilinguals who are less English dominant will perceive mid-vowel contrasts more robustly (e.g., closely to homeland speakers) than the bilinguals who are more dominant in English.

### 3.1 *Participants*

#### 3.1.1 Heritage Speakers of Brazilian Portuguese

A total of 12 HSs of BP participated in this study. Participants were eight females and four males ranging in age from 20 to 37 ($\bar{X}$ = 22.5; $s$ = 4.7). They lived in upstate New York when the data were collected. Seven participants

were students at an American university, while the other five participants had already concluded their undergraduate studies. All of them had at least one parent who was born in Brazil. None of them reported having any speech or hearing disorder. In order to obtain demographic and background information about HSs, they were asked to answer a modified version of the *Questionnaire in English for Bilingual Speakers* (Montrul, 2012). A summary of the participants' responses about their language background is shown in Table 2.

In order to assess language dominance, participants were asked to complete the *Bilingual Language Profile: English-Portuguese* (BLP) (Birdsong, Gertken, & Amengual, 2012). Figure 1 shows the BLP results for each participant. Overall, participants showed a broad spectrum of Portuguese dominance (scores

TABLE 2 Language background and demographics reported by the HSs of BP

| Participant | Age | Gender | Country born (age moved to US) | Parent born in Brazil (state) | Language hearing at home before 5 years old | Age exposed to English (in years) | Age exposed to Portuguese (in years) | Formal Portuguese at school (in semesters) |
|---|---|---|---|---|---|---|---|---|
| 1 | 24 | F | USA | F (RJ) | E | At birth | At birth | Just started |
| 2 | 29 | M | USA | M (SP), F (MG) | P | 4–5 | At birth | 2 |
| 3 | 37 | M | Brazil (4) | M (SP) | P | 3 | At birth | 0 |
| 4 | 20 | F | USA | F (RJ) | E | 1 | Limited | Just started |
| 5 | 20 | M | USA | M (MG), F (MG) | P | At birth | At birth | Just started |
| 6 | 21 | F | Brazil (2) | M (RJ), F (RJ) | P | At birth | At birth | Just started |
| 7 | 22 | F | Japan (12) | M (SC) | P, J | 12 | At birth | Just started |
| 8 | 24 | F | USA | M (MG) | P, E, S | At birth | At birth | 1 |
| 9 | 20 | F | USA | F (RJ) | P, E | At birth | At birth | Just started |
| 10 | 20 | M | USA | M (SP) | P, E | At birth | At birth | 2 |
| 11 | 22 | F | USA | M (RJ) | P, E, S | 1 | At birth | 1 |
| 12 | 21 | F | USA | F (MG) | P | 7 | At birth | 2 |

*Note*: M = mother; F = father; E = English; P = Portuguese; S = Spanish; J = Japanese; RJ = Rio de Janeiro; MG = Minas Gerais; SP = São Paulo; SC = Santa Catarina.

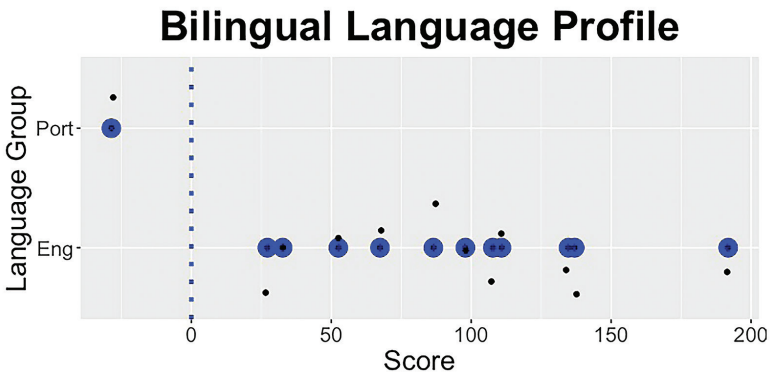


FIGURE 1 Language dominance of each HS of BP as a function of language group (English, Portuguese) as revealed by the BLP (Birdsong et al., 2012)
*Note*. The small black dots represent each HS in reference to her/his respective score. The blue dots represent the number of participants in relation to their language dominance.

ranged from −28.59 to +155.54; point difference = 184.13), whereas English scores exhibited a narrower range (scores ranged from +86.72 to +207.86; point difference = 121.14).

Only one participant obtained a negative score (Participant 7), showing that Portuguese is her/his dominant language. All the other HSs of BP are English dominant. Since studies on HLs have consistently shown the pervasive effects that language dominance can have on bilinguals' perception and production of mid vowels (e.g., Amengual, 2015, 2016a; Amengual & Chamorro, 2015; Sebastián-Gallés & Soto-Faraco, 1999), Participant 7 was excluded from the analysis. Therefore, a total number of 11 participants were considered for analysis from that point on.

From the language questionnaire (Montrul, 2012), additional information was obtained from participants. They were asked to rate their Portuguese language skills (*speaking, understanding, reading, writing*) from 0 (not good) to 6 (very good). Participants reported *understanding* as being their best skill ($\bar{X}$ = 4.09; $s$ = 1.81), followed by *reading* ($\bar{X}$ = 3.81; $s$ = 1.72), then *speaking* ($\bar{X}$ = 3.0; $s$ = 1.84), and that their weakest skill is *writing* ($\bar{X}$ = 2.54; $s$ = 1.50). In addition, participants rated their English language knowledge as better than their Portuguese knowledge in all four skills (see Figure 2). This result is similar to other studies on HLs in which HSs tend to rate their writing as their least developed and their listening as their most developed skill, while also rating their HL skills overall as lower than their skills in the majority language (e.g., Carreira & Kagan, 2011; Polinsky & Kagan, 2007).

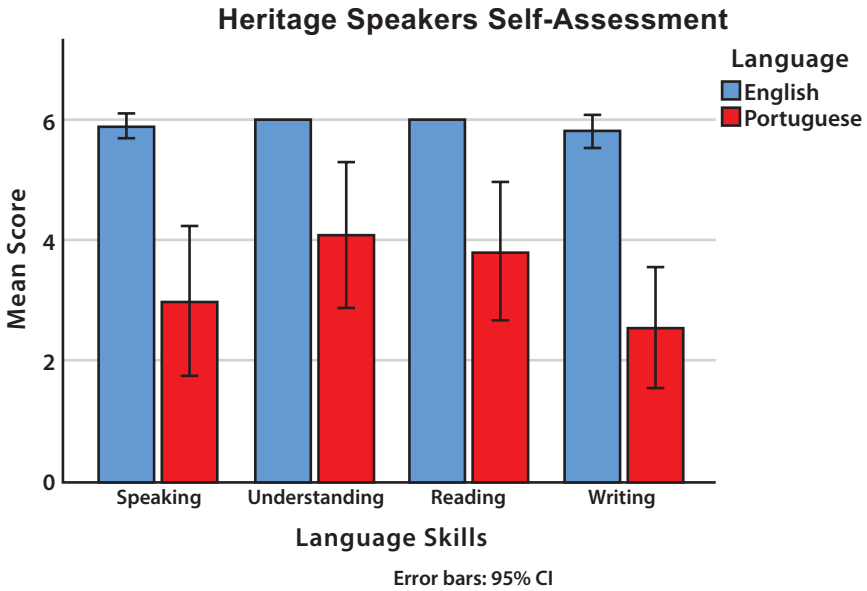


FIGURE 2 Self-assessment of English and Portuguese languages by HSs of BP
*Note*. On the y-axis, 0 means 'not good' and 6 means 'very good.' Error bars represent the 95% confidence interval for each mean.

The responses on the language questionnaire background were grouped into two categories: use of Portuguese and personal connections with Portuguese (language and culture). Heritage speakers provided information about their use of Portuguese by furnishing the average percentage per week Portuguese use with friends and family, at school, and when talking to themselves. The use of Portuguese was calculated for each participant in the following way: The percentages provided by HSs were added and the overall percentage was obtained (e.g., If the sum of the four questions was 70%, the relative overall percentage over the four questions was 17.5%). The percentage was then converted to decimals (e.g., 17.5% corresponds to 0.175). Information about HSs' personal connections with Portuguese was obtained through participants' ratings of four statements about language and culture from 0 (disagree) to 6 (agree). The four statements were: *I feel like myself when I speak Portuguese; I identify with a Portuguese-speaking culture; It is important to use Portuguese like a native speaker; I want to think I am a native speaker of Portuguese*. Participants' rates were added and their overall percentages were transformed into decimals (e.g., a total of 16 points corresponded to an overall percentage of 66.67%, or 0.667). Participants also provided self-assessments of their Portuguese skills (i.e., *reading, speaking, understanding, writing*) on a scale from 1 (not good) to 6 (very

good). For the analyses of their Portuguese skills, scores for each skill were considered separately to see whether Portuguese language skills correlated among themselves.

A series of correlations between self-assessment of Portuguese skills, use of Portuguese, connections with Portuguese language/culture, and scores from BLP were conducted using Pearson correlation in R (R Development Core Team, 2017), and interpretation of effect size (Cohen, 1988): $r = 0.10$ (small), $r = 0.30$ (medium), $r = 0.50$ (large). The correlation results, along with their significance levels, are shown in Table 3. The results showed significant negative correlations among scores in the BLP and all the other variables; all correlations had large negative effect sizes (*use* $r = -0.77$, $p < 0.01$; *connections* $r = -0.72$, $p < 0.05$; *speaking* $r = -0.81$, $p < 0.01$; *understanding* $r = -0.92$, $p < 0.001$; *reading* $r = -0.77$, $p < 0.01$; *writing* $r = -0.61$, $p < 0.05$). These results show that the lower their scores were on the BLP, the higher HSs rated their Portuguese skills, the more they used Portuguese in their lives, and the more they felt connected with Portuguese language and culture.

The range of HSs' self-declared proficiency in Portuguese was very broad. Considering their overall skills, on a scale of 1 to 6, their overall proficiency ranged from 1.25 (the lowest grade) to 5.5 (the highest grade). This variation in HL proficiency has also been observed in a number of studies that show that HSs tend to have a wide range of experiences with their HL (e.g., Montrul, 2016; Polinsky, 2018). The four Portuguese language skills correlated significantly among each other (*speaking* [*understanding* $r = 0.89$, $p < 0.001$; *reading* $r = 0.88$, $p < 0.001$; *writing* $r = 0.79$, $p < 0.001$]; *understanding* [*reading* $r = 0.93$,

TABLE 3 Correlations between language and background information provided by HSs of BP. Correlation coefficients come from Pearson correlation. Significance levels: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

| | Use | Connection | Speaking | Understanding | Reading | Writing | Scores in BLP |
|---|---|---|---|---|---|---|---|
| Use | 1 | 0.27 | 0.45 | 0.58 | 0.47 | 0.46 | $-0.77^{**}$ |
| Connection | | 1 | $0.68^{*}$ | $0.78^{**}$ | $0.64^{*}$ | 0.44 | $-0.72^{*}$ |
| Speaking | | | 1 | $0.89^{***}$ | $0.88^{***}$ | $0.79^{**}$ | $-0.81^{**}$ |
| Understanding | | | | 1 | $0.93^{***}$ | $0.71^{*}$ | $-0.92^{***}$ |
| Reading | | | | | 1 | $0.77^{**}$ | $-0.77^{**}$ |
| Writing | | | | | | 1 | $-0.61^{*}$ |
| Scores in BLP | | | | | | | 1 |

$p < 0.001$; *writing* $r = 0.71$, $p < 0.05$]; *reading* [*writing* $r = 0.77$, $p < 0.01$]), indicating that those HSs who self-evaluated one Portuguese skill more positively also evaluated the other skills positively. Personal connections with the language and culture were significantly (and positively) correlated with *speaking* ($r = 0.68$, $p < 0.05$), *understanding* ($r = 0.78$, $p < 0.01$) and *reading* ($r = 0.64$, $p < 0.05$), but were not significantly correlated to *writing* ($p > 0.05$). It is important to mention that *writing* (i.e., productive mode) is a skill acquired mainly at school (but notice that one of the participants became literate in Portuguese through home schooling). Therefore, the lack of significant correlation between *writing* and personal connections may not come as a surprise. As Montrul (2016) wrote, "heritage speakers display asymmetries in the productive and receptive modes and in oral and written language" (p. 44).

Surprisingly, the use of Portuguese did not significantly correlate with personal connections ($p > 0.05$) or with any of the four Portuguese skills ($p > 0.05$); that is, whether HSs use Portuguese frequently or rarely in their lives did not correlate with their skills in Portuguese or with how they felt about the language/culture. One might expect that the use of Portuguese would correlate with language skills since L2 use by early bilinguals has been shown to influence their L2 language skills (e.g., Derwing, Munro, & Thompson, 2008), such as in the perception and production of mid vowels by Catalan-Spanish bilinguals (e.g., Mora, Keidel, & Flege, 2015; Mora & Nadeu, 2012). A closer look at the data shows that participants used Portuguese much more often with their families at home than in other contexts. On average per week, HSs use Portuguese 51% of the time when speaking with their families, whereas only 15% and 9% when they are at work/school or with friends, respectively. Therefore, their overall use of Portuguese was mainly restricted to their homes, showing a "family connection to their home language" (Boon & Polinsky, 2014, p. 3).

Gertken, Amengual, and Birdsong (2014) conducted a comprehensive discussion on what it means to say that someone is dominant in a given language. They argue that, even though language dominance is often related to proficiency, these two concepts (i.e., language dominance and proficiency) should be considered separately because other factors are involved in language dominance, such as cultural identification and motivation. The authors also state that "bilinguals are not necessarily dominant in one language across the board" (p. 212), and that aspects of language dominance are gradient and not dichotomous. The correlations in the present study point in this direction; that is, as HSs become less dominant in English, their Portuguese skills, use of the language, and personal connections with Portuguese increase. Taking into consideration that language dominance has been considered a relative term, and that "discrete classifications of language dominance can obscure

rich data about variation within groups" (p. 213), considerations about levels of gradation in English language dominance were considered in this study. On the BLP, HSs' lowest score was 27.154, their highest was 191.788, the median was 97.9, and the mean was 95. Based on this information, a cut-off criterion of 95 was used, which divided HSs into two groups according to their scores on the BLP: a group (with scores below 95) with less dominance in English (i.e., lower English dominant, LED, $n = 5$) and another group (with scores above 95) with more dominance in English (i.e., higher English dominant, HED, $n = 6$).

The use of Portuguese, personal connections with the language, and the Portuguese skills of each group were compared in order to know whether they would differ as functions of degrees of English dominance. In terms of their scores on the BLP, HED showed strong dominance in English ($\bar{X} = 129.98$, $s = 34.01$, *range* = 97.9–191.78), whereas LED showed weaker language dominance in English ($\bar{X} = 53.219$, *s.* = 24.54, *range* = 27.14–86.46). In terms of Portuguese skills, HED rated their four Portuguese language skills much lower (*speaking:* $\bar{X} = 1.6$, $s = 1.21$, *range* = 4–1; *understanding:* $\bar{X} = 2.8$, $s = 1.47$, *range* = 5–1; *reading:* $\bar{X} = 2.6$, $s = 1.50$, *range* = 5–1; *writing:* $\bar{X} = 1.8$, $s = 1.47$, *range* = 4–1) than LED (*speaking:* $\bar{X} = 4.6$, $s = 0.80$, *range* = 6–4; *understanding:* $\bar{X} = 5.6$, $s = 0.54$, *range* = 5–6; *reading:* $\bar{X} = 5.2$, $s = 0.44$, *range* = 5: *writing:* $\bar{X} = 3.4$, $s = 1.14$, *range* = 5–2). These two groups also differentiate themselves in terms of connections with Portuguese; that is, HED reported having less personal connection with the language/culture ($\bar{X} = 0.61$, $s = 0.28$, *range* = 0.12–0.87) than LED ($\bar{X} = 0.90$, $s = 0.18$, *range* = 0.70–0.91). They also differed in terms of use of the HL; that is, HDE used Portuguese less often ($\bar{X} = 0.1$, $s = 0.07$, *range* = 0–0.17) than LED ($\bar{X} = 0.23$, $s = 0.12$, *range* = 0.05–0.35).

A series of Mann-Whitney U Tests showed that, in fact, HDE and LDE differed significantly in their self-assessment of the following Portuguese skills: *speaking* ($W = 28.5$, $p < 0.05$), *understanding* ($W = 29$, $p < 0.05$), and *reading* ($W = 28$, $p < 0.05$). These two language groups did not differ significantly in terms of *writing* and *use* of Portuguese ($p > 0.05$), and *connections* was of borderline significance ($p = 0.0530$). Therefore, for this group of HSs, degrees of language dominance are associated significantly with the Portuguese *speaking, understanding*, and *reading*, but not with *writing, use* or *connections with the language* (information obtained from the HSs' self-reported language background questionnaire). As Figure 3 shows, there is no overlap in the distribution of BLP scores between HED and LED for Portuguese. For English, there is some overlap between the groups; however, the group mean and the median for English are different for HED and LED, indicating that they are different groups in terms of their English score. There is also a greater distance between the scores in Portuguese and English for HED than for LED.

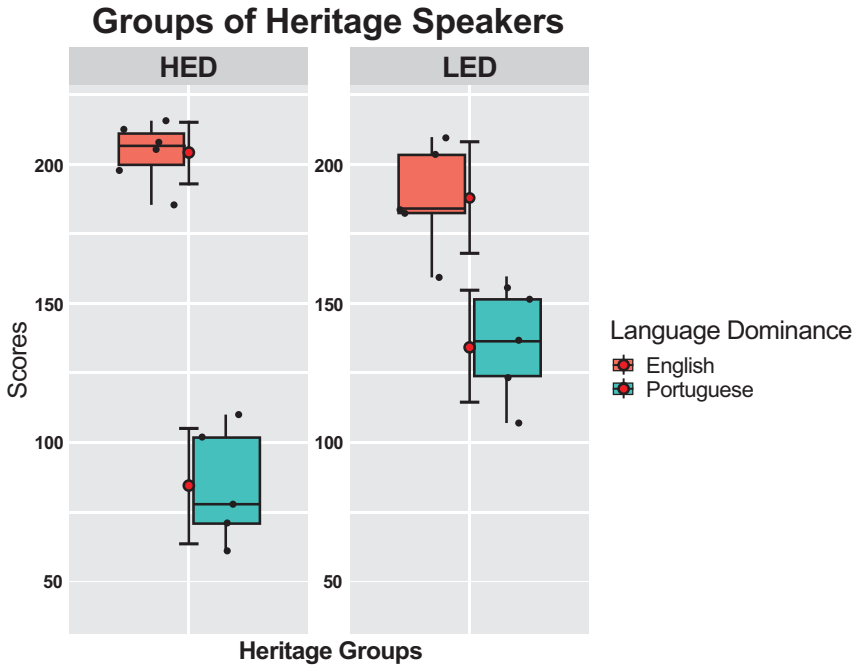


FIGURE 3 Distribution of the scores in English and in Portuguese from the BLP (Birdsong et al., 2012) by HSS of BP across two language groups: HED and LED
*Note.* Red dots correspond to group mean, bars inside the boxes correspond to the median, whiskers and boxes represent the range of responses, with the end of the whiskers representing the lowest and the highest scores, indicating variability outside the lower and upper quartile. Error bars represent standard error of the mean. Normality of scores was tested visually through a Q-Q plot and statistically through the Shapiro-Wilk test ($p > 0.05$).

### 3.1.2 Homeland Speakers

A total of 14 homeland speakers (HomeS), born in Brazil, participated in this study (10 females, four males). They were living in a city located in the southeast of the state of Minas Gerais (Brazil), where the data were collected. Their ages ranged from 29 to 68 years old ($\bar{X} = 50.5$; $s = 14.6$). They all reported having at least a high school degree, as well as normal speech, hearing, and vision. This group of HomeS was included to provide a baseline for comparisons in the perception experiments.

## 3.2 *Stimulus Materials, Equipment and General Procedures*

The stimulus material used in the experiments was produced by a female native speaker of BP (47 years old) whose L2 is English, and who had lived in

the United States for 20 years at the time of the recordings. The speaker was recorded in a quiet room in a language lab at an American university. Care was taken to make sure acoustically that the stimuli were more Portuguese-like than English-like, for instance, by producing short-lag values for Portuguese VOT and monophthongs (e.g., [e] and [o]), instead of diphthongs (e.g., [eɪ] and [ow]), as would happen in English (see examples in Figure 4).[1]

The material used in the experiments consisted of minimal pairs with Portuguese mid-vowel contrasts and minimal pairs with non-target vowels. Target sounds were always in stressed position of real words. Most of the words were bisyllabic, but a few were monosyllabic, with all being produced in the following carrier phrase: [target word] *é a palavra*. ('____ is the word.'). Natural sounds were manipulated in a way that only the target vowel would change in the stimulus word; that is, the adjacent phonetic segments remained the same throughout the stimuli. Acoustic manipulations were done in Praat (Boersma & Weenink, 2014). Sounds were normalized by intensity at 75 dB through a Praat script. Details on the stimuli used in each experiment are given in their respective sections.

Data from HSS were collected in a quiet room at an American university. Data from HomeS were collected in Brazil (southeast of Minas Gerais), in a quiet room at the researcher's home or at the participant's house. The researcher explained the experiments to the participants in Portuguese. Once participants started the experiments, they would hear the instructions again in Portuguese. After participants finished the experiments, they answered their respective language background questionnaires in printed form. Specific procedures for each experiment are given in their respective sections. The experiments were presented in Psychopy (Peirce & MacAskill, 2018) using the researcher's computer (MacBook Air, 2017), and a set of Sennheiser HD 206 closed dynamic headphones was used, through which participants adjusted the volume to their comfort. Their participation took around 30 minutes.

## 4 Experiment 1: AXB Discrimination Test

The first experiment, an AXB discrimination test, investigated whether HSS were able to discriminate Portuguese mid vowels categorically. The scores HSS obtained in this test informed us of the extent to which HSS have learned to perceptually discriminate Portuguese vowel contrasts, and whether degree of language dominance in English (higher or lower) had an effect on their perception. It is hypothesized that HSS who are more English dominant (i.e., HED)

will have more difficulties in discriminating vowel contrasts since degree of language dominance has been shown to affect vowel perception in the minority language (e.g., Amengual, 2015; Gertken, Amengual, & Birdsong, 2014). It is also hypothesized that HSs may demonstrate more discrimination difficulties than HomeS since HSs form a heterogenous group that has been exposed to limited input (e.g., Polinsky & Scontras, 2019).

### 4.1 *Material Stimuli and Method for Experiment 1*

Participants heard three consecutive words and were asked to choose a word that sounded like the word in the middle (X), with their options being either the first word (A) or the third word (B). When choosing the first word, they had to click on the left arrow on the computer keyboard, and when choosing the third word, they had to click on the right arrow. The items (A) and (B) were always different from each other, and one item was always the correct response (see Figure 5).

All possible AXB arrangements were used in the experiment: AAB, ABB, BAA, and BAB. The stimulus triads were counterbalanced in their presentation, randomized, and presented in one block. Each stimulus triad was presented only once. The inter-stimulus interval was 1.0 second, and the inter-trial interval was 2.7 seconds from the onset of the triad. There was a training section (8 triads) and stimulus triads as distractors (56 triads), all of which were non-target minimal pairs. Participants would move from one triad to the next only after they had clicked their response. The target words used in Experiment 1 are shown in Table 4. There were seven minimal pairs for each vowel contrast (i.e., front and back).

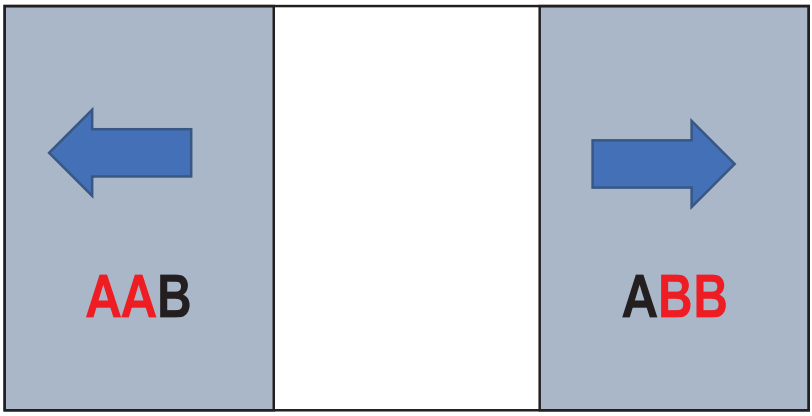


FIGURE 5 Screen shot of the computer interface in Experiment 1

TABLE 4 List of the Portuguese target words containing minimal pairs with /ɛ/-/e/ and /ɔ/-/o/ contrasts used in the AXB discrimination test (Experiment 1), along with their respective translation into English

| **/e/-/ɛ/** |
|---|
| g/e/lo 'ice' – g/ɛ/lo 'I freeze' |
| p/e/ 'the letter p' – p/ɛ/ 'foot' |
| s/e/de 'thirst' – s/ɛ/de 'headquarters' |
| m/e/ta 'get in' – m/ɛ/ta 'goal' |
| s/e/lo 'stamp' – s/ɛ/lo 'I stamp' |
| p/e/so 'weight' – p/ɛ/so 'I weigh' |
| s/e/co 'dry' – s/ɛ/co 'I dry' |
| **/o/-/ɔ/** |
| g/o/sto 'taste' – g/ɔ/sto 'I like' |
| /o/lho 'eye' – /ɔ/lho 'I look' |
| p/o/ço 'well' – p/ɔ/sso 'I can' |
| p/o/de 'you could' – p/ɔ/de 'you can' |
| j/o/go 'game' – j/ɔ/go 'I play' |
| av/o/ 'grandfather' – av/ɔ/ 'grandmother' |
| r/o/xa 'purple' – r/ɔ/cha 'rock' |

### 4.2 *Results of Experiment 1*

Experiment 1 was conducted to verify whether HSS of BP were able to discriminate Portuguese mid vowels categorically. There was a total of 1,456 target responses (i.e., 26 participants × 14 minimal pairs × 4 combinations; the distribution of responses can be seen in Figure 6).[1] There is virtually no difference between the language groups. The three language groups (i.e., LED, HED, HomeS) successfully discriminate the vowels, reaching ceiling effects for both vowel contrasts: mid-front vowel contrast (HED: $\bar{X} = 0.98$; $s = 0.11$; LED: $\bar{X} = 0.98$; $s = 0.11$; HomeS: $\bar{X} = 0.99$; $s = 0.09$) and mid-back vowel contrast (HED: $\bar{X} = 0.97$; $s = 0.12$; LED: $\bar{X} = 0.99$; $s = 0.11$; HomeS: $\bar{X} = 0.98$; $s = 0.09$).

Generalized linear mixed-effect models (GLMM), through *glmer*() and with the binomial link function from *lme4* package in R (R Development Core Team, 2017) were used (e.g., Casillas & Simonet, 2016). The statistical analyses included the response variables of language group (i.e., LED, HED,

HomeS) and target vowel contrast (i.e., mid-front, mid-back) as fixed factors, and intercepts for participants as random factors. The results of the overall mixed-effects logistic regression models and the summary of model comparisons are shown in Appendices 1 and 2,[1] respectively. There was no significant interaction between language group and mid-vowel contrast ($\chi(2) = 0.938$, $p > 0.05$), no main effect of language group ($\chi(2) = 0.779$, $p > 0.05$), and no main effect of target phoneme contrast ($\chi(1) = 0.256$, $p > 0.05$). We can conclude that, independent of the degree of English dominance, HSs of BP were able to distinguish Portuguese mid-vowel contrasts successfully. In addition, no difference in discriminating vowel contrasts between HSs and HomeS was found, indicating that HSs and HomeS discriminate both vowel contrasts at the same level in this test.

## 5 Experiment 2: 2AFC Identification Test

The second perception experiment, a two alternative-forced choice (2AFC) identification test, provided additional information regarding HSs' perceptual categorization by investigating whether participants differentiated Portuguese mid-vowel contrasts when they were presented along an acoustic continuum going from low to high mid vowel. Along the lines of the first experiment, for this second experiment, it is hypothesized that HSs will have more difficulty than HomeS since HSs have had varied experiences with Portuguese. It is also hypothesized that the level of difficulty will be related to the degree of English dominance such that those HSs who are less dominant in English will do better than those who are more dominant in English (Amengual, 2015; Gertken, Amengual, & Birdsong, 2014).

### 5.1 *Material Stimuli for Experiment 2*

Following Amengual and Chamorro (2015), the stimuli consisted of seven steps on a continuum for each vowel contrast (i.e., /ɛ/-/e/ and /ɔ/-/o/). The target vowels were sliced from the words */pɛ/* ('foot'), */pe/* (letter 'p'), */avɔ/* ('grandmother'), and */avo/* ('grandfather') to be synthesized (the speaker was the same speaker who produced the material for Experiment 1). It was important to create sets of sounds that were as natural as possible; that is, moving from one step to the next needed be smooth and natural, while avoiding metallic or robotic sounds. To achieve this goal, several attempts were made using

1 The reader can access Appendices 1 to 4, Figures 4 and 6, and Tables 6, 7, 8, 10 and 11 at OSF Open Science Framework via the following link: https://osf.io/5fybh/?view_only=95037c2293024610925a8d6d18a8d88f.

different scripts. The best sets of sounds (i.e., those that sounded more natural to the researcher) were produced by a Praat script created by Winn (2016). The formant frequencies F1 and F2 of the original sounds were synthetized into seven step continua for each pair of mid vowels in roughly equidistant increments. For the front vowels, F1 linearly decreased from 706.6 hertz (Hz) to 434.5 Hz, with a mean decrease of –45.35 Hz between steps (range: –38.9 to –52.4), and F2 linearly increased from 2,294.4 Hz to 2,581.3 Hz, with a mean increase of +47.81 Hz between steps (range: 39.6 to 55.8). For the back vowels, F1 linearly decreased from 556.0 Hz to 413.7 Hz, with a mean decrease of –23.71 Hz between steps (range: –18.3 to –30.5), and F2 linearly decreased from 954.2 Hz to 852.9 Hz, with a mean decrease of –16.88 Hz between steps (range: –11.7 to –20.7). Other parameters, such as duration and intensity, were kept the same throughout the continua. Figure 7 shows the seven vowel continua for each contrast in the vowel space and Table 5 shows the list of F1 and F2 values of each vowel. The seven-step continua were normalized for peak intensity, which scales all the token sounds to the average intensity level of 75dB. The seven steps were then inserted in */p/* and /av/ such that the only change in the sets of words were the target vowels.

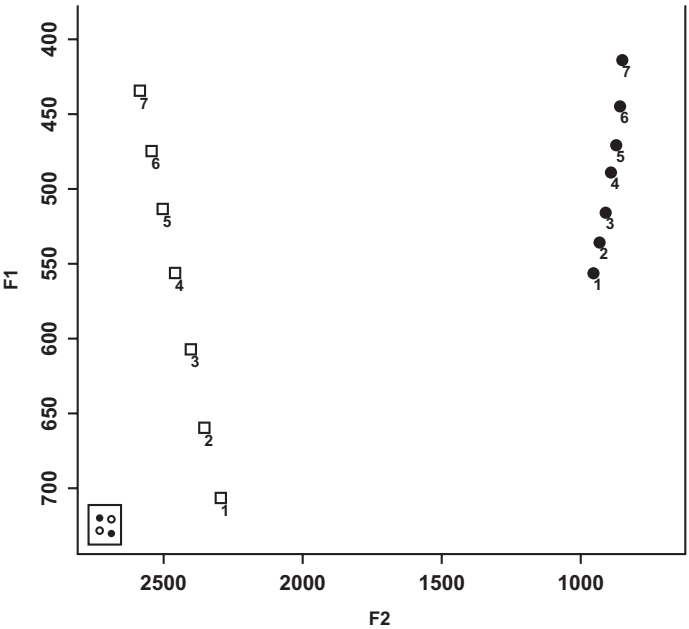


FIGURE 7 Synthesized seven-step vowel continua for /ɛ/-/e/ (on the left side of the chart) and /ɔ/-/o/ (on the right side of the chart)
*Note*. Non-normalized vowel plot created in the NORM website (Thomas & Kendall, 2007). Formant values are in Hz.

TABLE 5 Table with F1 and F2 values for each step of the continua used as stimuli in Experiment 2

| | Mid-Front Vowels | | Mid-Back Vowels | |
|---|---|---|---|---|
| **Steps** | **F1 (Hz)** | **F2 (Hz)** | **F1 (Hz)** | **F2 (Hz)** |
| 1 | 706.6 | 2,294.4 | 556.0 | 954.2 |
| 2 | 659.4 | 2,350.2 | 534.8 | 934.2 |
| 3 | 607.0 | 2,398.8 | 514.6 | 913.5 |
| 4 | 556.7 | 2,453.5 | 488.7 | 895.1 |
| 5 | 513.0 | 2,498.6 | 470.4 | 877.8 |
| 6 | 474.1 | 2,538.2 | 444.2 | 864.6 |
| 7 | 434.5 | 2,581.3 | 413.7 | 852.9 |

### 5.2 *Method for Experiment 2*

Experiment 2 was divided into two consecutive blocks. In the first block, listeners heard variations of the /ɔ/-/o/ continuum, and in the second block, listeners heard variations of the /ɛ/-/e/ continuum. In each block, participants saw two drawings representing the target words on the screen. For the mid-back vowel contrasts, participants saw a picture of a grandmother */avɔ/* on the left side of the screen and a picture of a grandfather */avo/* on the right side of the screen. For the mid-front vowel contrast, participants saw a picture of a foot */pɛ/* on the left side of the screen and the letter 'p' */pe/* on the right side of the screen. The pictures were all black and white, similar in size, and oriented the same way on each screen (see Figure 8). Participants were asked to match the word they heard with the picture they saw on the screen.

The order of presentation of the stimuli in each block was randomized. The experiment was self-paced; that is, the next stimulus was only presented after the participant had selected a response. During the test, each vowel step was presented 10 times, resulting in a total of 140 items for each block. There was a practice section in which listeners heard the randomized seven steps for mid-back and mid-front vowel contrasts in their respective sections. No distractors were added.

### 5.3 *Results of Experiment 2*

To make ensure participants understood the test were answering the test as expected (i.e., choosing between low or high mid vowels), a criterion for a performance level of 50% (0.5 points) or more was implemented in the following

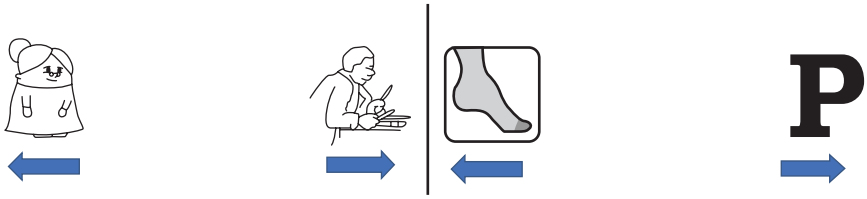


FIGURE 8 Screen shots of the computer interface for Experiment 2
*Note*. The pictures on the left side represent words with mid-back vowels, whereas the pictures on the right side represent words with mid-front vowels.

way: The difference between the perception of low vowel responses (i.e., /ɛ/ and /ɔ/) at both endpoints of the continuum was obtained (i.e., steps 1–3 and steps 5–7, not including midpoint step 4), and then one endpoint was subtracted from the other. Only participants (i.e., HSs and HomeS) who had a difference of 50% or more (0.5 points) between these two endpoints were included in the analyses. Therefore, the desired target of correct perception of low-mid vowels was 50% or higher. In this manner, participants who were answering the test randomly were separated from those who were able to answer the test correctly; for instance, if a participant perceived /ɛ/ 80% of the time in steps 1–3 (i.e., 24 out of 30) and 33.33% of the time in steps 5–7 (i.e., 10 out of 30), the difference was 46.67%, and (s)he was excluded from the analyses. The results showed that all HomeS performed above the cut-off criterion; for the perception of /ɛ/, their differences ranged from 86.67% to 100%, and for the perception of /ɔ/, their differences ranged from 73.33% to 100%. In the HS group, two participants were excluded from the test; one obtained −66.7% for /ɔ/ and another obtained −86.67 for /ɛ/, where the negative percentages showed that they were choosing the vowels in the opposite direction (i.e., outliers). These two participants belonged to the HED group. The remaining HSs performed at or above the cut-off criterion; for the perception of /ɛ/, HSs' differences ranged from 86.66% to 100%, and for the perception of /ɔ/, their differences ranged from 50% to 100%. Therefore, the total participants considered in the analyses of Experiment 2 was 23 (HED, $n$ = 4; LED, $n$ = 5; HomeS, $n$ = 14).

The 2AFC identification test informed us whether participants showed category boundaries for each vowel realization on a continuum through sensitivity to acoustic correlates for mid vowels: as front vowels went from /ɛ/ to /e/, F1 decreased and F2 increased, and as back vowels went from /ɔ/ to /o/, both F1 and F2 decreased. This test also informed us whether their place of category boundary (i.e., crossover point) varied as a function of language dominance. It was expected that the perceptual space would shift somewhere on the

continuum, and that a small area of ambiguity would occur, representing the area where listeners may not be sure which sound they were hearing, meaning both categories were possible. There was a total of 3,220 responses (i.e., 23 participants × 7 steps × 2 mid-vowel contrasts × 10 repetitions). The results for each vowel contrast are shown separately in the following sections.

### 5.3.1 Mid-Front Vowels

The distribution responses as a function of steps for each language group in the identification of the /ɛ/-/e/ contrast is shown in Figure 9. The percentage of each vowel identified by language group is shown in Table 6.[1] The percentage among language groups suggested that, overall, LED and HomeS tended to choose /ɛ/ more often than /e/ on the continuum; however, a visual inspection of the graphs (Figure 9) suggests that language groups overlapped in almost all steps, except for step 4, which corresponds to the ambiguous area (the middle point between /ɛ/ and /e/). Therefore, the percentage differences presented in Table 6 may be attributed to differences in step 4.

To verify whether there were differences in the slopes among groups, the slope of fitted lines (i.e., the sharpness of the boundary for each language group) for the vowel contrast was estimated and compared via the function

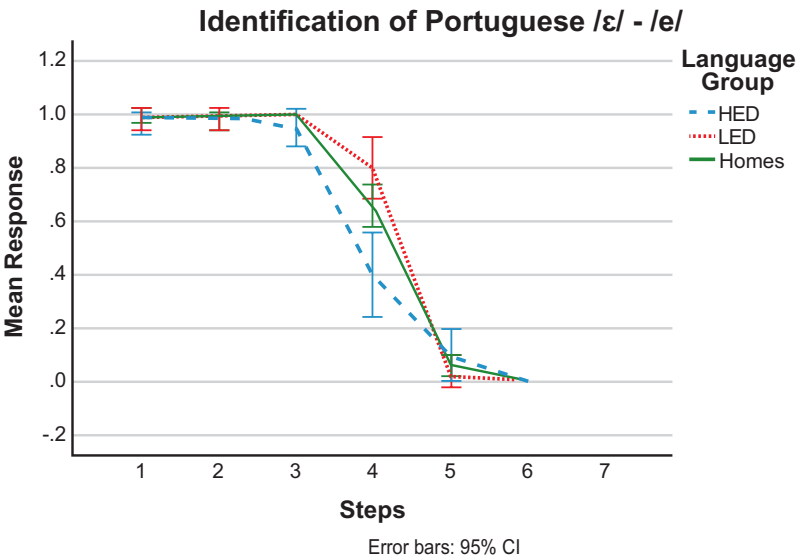


FIGURE 9 Overall distribution of the perception of Portuguese mid-front vowels on a continuum (Experiment 2)
*Note.* Mean responses are provided based on language group and vowel step from low- to high-mid vowels. On the y-axis, the number 1 corresponds to /ɛ/ (pé 'foot') whereas 0 corresponds to /e/ (pê 'letter p').

lstrends (Lenth, 2014) in R (R Development Core Team, 2017) by fitting a logistic regression model that included steps and language groups. The estimate slopes were −2.3222 for HED, −3.1548 for LED, and −2.9322 for HomeS. The difference in slopes among language groups was non-significant ($p > 0.05$), as we expected from Fig. 9. (see results of the slopes in Table 7).[1] Therefore, all the groups (i.e., HED, LED, HomeS) labeled the vowels on the continuum in a similar way.

The crossover point (i.e., category boundary location) is the point at which the shift from one category to another occurs on a continuum. Crossover points were calculated through logistic regression using the *cross_over* () function from the package *lingStuff* in an R script (Casillas, 2018). Crossover points in the identification of the Portuguese /ɛ/-/e/ contrast, with their respective values for each language group, are shown in Figure 10, where we see that HED tended to shift boundaries earlier and LED tended to shift boundaries later than the other groups, although this difference is small. The crossover point for HED was 3.925, the one for LED was 4.272, and the one for HomeS was

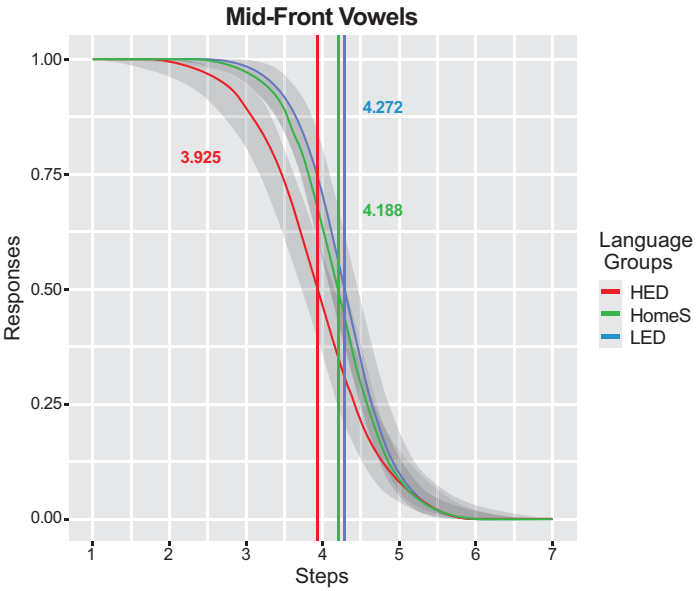


FIGURE 10 Curves representing the logistic regression model with group-based average slopes
*Note*. The vertical bars represent the phonetic boundary crossover point for each language group. Homeland speakers are represented in green, the HED group is represented in red, and the LED group is represented in blue. The prediction bands (95% confidence) indicate residual variance. On the y-axis, the number 1 corresponds to /ɛ/'s end point and 0 corresponds to /e/'s end point.

4.188. The results of the logistic regression with crossover points for the /ɛ/-/e/ contrast is in Appendix 3.[1] No statistical difference was verified among groups ($p > 0.05$). All groups shifted boundaries at the same location, around step 4.

In sum, the analyses showed that HED and LED perceived the mid-front vowel contrast categorically, and that degree of English dominance (i.e., higher vs. lower) did not affect how HSs identified mid-front vowels in Portuguese. The analyses also showed that HSs' responses did not differ from those of HomeS, indicating that differences in experience with Portuguese (e.g., HSs had limited input) did not prevent HSs from successfully identifying the /ɛ/-/e/ contrast.

### 5.3.2 Mid-Back Vowels

The distribution responses as a function of steps for each language group in the identification of the mid-back vowel contrast are shown in Figure 11. Percentages of responses for each vowel by language group are shown in Table 8.[1] In terms of overall percentage difference (see Table 8), LED showed a tendency to choose more /ɔ/ than /o/ on the continuum, whereas the other groups did not show any bias toward either high or low vowels. Figure 11 shows that, contrary to the /ɛ/-/e/ contrast, the identification of /ɔ/-/o/ showed some variability, especially regarding HED. Whereas HSs who were LED and HomeS overlapped in almost all steps, HSs who were HED showed a slope that differentiated it from the other groups; that is, HED's slope is less sharp and its change is smooth, whereas the slopes of LED and HomeS change more abruptly from one category to another. Overall, HED indicated less confidence in labelling the /ɔ/-/o/ contrast throughout the vowel continuum (i.e., more centralized on the y-axis) when compared to the other groups.

First, the estimated slopes and comparisons among language groups were done by fitting a logistic regression model. The slope for HED was –1.19. the one for LED was –3.40, and the one for HomeS was –2.72. The results of comparisons among slopes are presented in Table 9.

HSs who were LED and HomeS had slopes that did not differentiate themselves significantly from each other ($p > 0.05$). HSs who were HED had a slope that was significantly different from those who were LED ($p < 0.001$) and from those who were HomeS ($p < 0.001$).

Since there were significant differences among language groups, subsequent analyses investigated where these differences were located (i.e., on which steps of the continuum). A logistic regression analysis was conducted, which included responses as the independent variable and language group (i.e., HED, LED, HomeS) and step (i.e., from 1 to 7) as fixed factors. A summary of the overall logistic regression model is given in Table 10.[1] As predicted, the statistical results for the /ɔ/-/o/ contrast revealed that the interaction between language

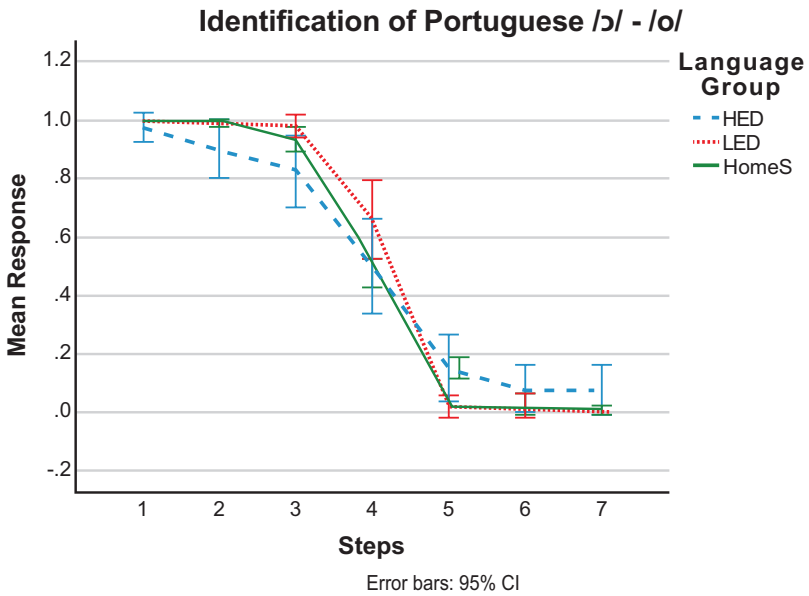


FIGURE 11 Overall distribution of the perception of Portuguese mid-back vowels on a continuum (Experiment 2)
*Note*. Mean responses are provided based on language group and vowel step from low- to high-mid vowels. On the y-axis, the number 1 corresponds to /ɔ/ (avó 'grandmother') whereas 0 corresponds to /o/ (avô 'grandfather').

TABLE 9 Results of the language pair comparisons of slopes for the identification of the Portuguese /ɔ/-/o/ contrast in Experiment 2. Model coefficient estimates, standard errors (SE), corresponding z-values, and *p*-values are included. *P*-values that are less than 0.05, 0.01, and 0.001 are flagged with one *, two **, and three *** stars, respectively.

| Pairs | Estimate difference in slopes | SE | z.ratio | p.value |
|---|---|---|---|---|
| HED-HomeS | 1.533 | 0.251 | 6.111 | *** |
| HED-LED | 2.214 | 0.524 | 4.221 | *** |
| HomeS-LED | 0.681 | 0.551 | 1.236 | $p > 0.05$ |

group and step was significant (*Deviance* = −42.312, $p < 0.001$), as was the main effect of step (*Deviance* = −1558.1; $p < 0.001$), but there was no main effect of language group (*Deviance* = −3.397; $p > 0.05$).

Since there was an interaction, pairwise comparisons were conducted among variables. The results of the comparisons are shown in Table 11.[1] Significant differences were found between HSs who were HED and HomeS at almost all steps: step 2 ($p < 0.05$), step 3 ($p < 0.05$), step 5 ($p < 0.01$), step 6 ($p < 0.05$), and step 7 ($p < 0.05$). Significant differences between HSs who were HED and LED were found at step 3 ($p < 0.05$). At the first endpoint (i.e., steps 2 and 3), HED heard significantly less instances of /ɔ/ ($\bar{X} = 0.862$) than HomeS ($\bar{X} = 0.964$). At the second endpoint (i.e., steps 5, 6 and 7), HED heard significantly more instances of /ɔ/ ($\bar{X} = 0.1$) than HomeS ($\bar{X} = 0.014$). The difference found between HED and LED was at step 3, where HED heard fewer instances of /ɔ/ ($\bar{X} = 0.825$) than LED ($\bar{X} = 0.98$).

These results indicate that HED was less categorical than HomeS (e.g., HED labeled more /ɔ/ and /o/ vowels where they were not expected). No difference among groups was found at step 1 (i.e., the extreme end point on the left side of the continuum) or step 4 (i.e., the ambiguous area). The results suggest that degrees of English dominance among HSs may affect the identification of the /ɔ/-/o/ contrast since HSs who were HED had more difficulties with mid-back vowel categorizations, whereas HSs who were LED were more successful. In addition, LED perceived the /ɔ/-/o/ contrast in a more homeland-like fashion since differences among LED and HomeS were not statistically significant, displaying similarities in their categorization.

Next, the crossover points in the identification of the Portuguese /ɔ/-/o/ contrast for each language group were calculated. Figure 12 shows the perceived category boundary for the groups. The LED group tended to shift boundaries a little later than the other groups, showing an overall tendency to hear more instances of /ɔ/. The shaded band is larger for HED than for the other language groups, showing more variability. The crossover point for LED was 4.170, the one for HED was 4.0, and the one for HomeS was 3.985. The results of the logistic regression with crossover points for the /ɔ/-/o/ contrast are in Appendix 4.[1] No statistical differences were found among language groups ($p > 0.05$), indicating that all groups shifted boundaries at the same location, around step 4.

In sum, degree of English dominance among HSs of BP was shown to affect HSs identification of the Portuguese /ɔ/-/o/ contrast, but not for the /ɛ/-/e/ contrast. For the mid-back vowels, HSs who were HED demonstrated less consistency and less confidence in their identification than HSs who were LED (e.g., HED displayed a slope that was significantly less sharp than those of LED and HomeS). The HED group was significantly different from HomeS at almost all steps, whereas it differed from LED only at step 3. This result indicates that the mid-back vowel contrast is more difficult for HSs to perceive, but only for those HSs who were highly English dominant.

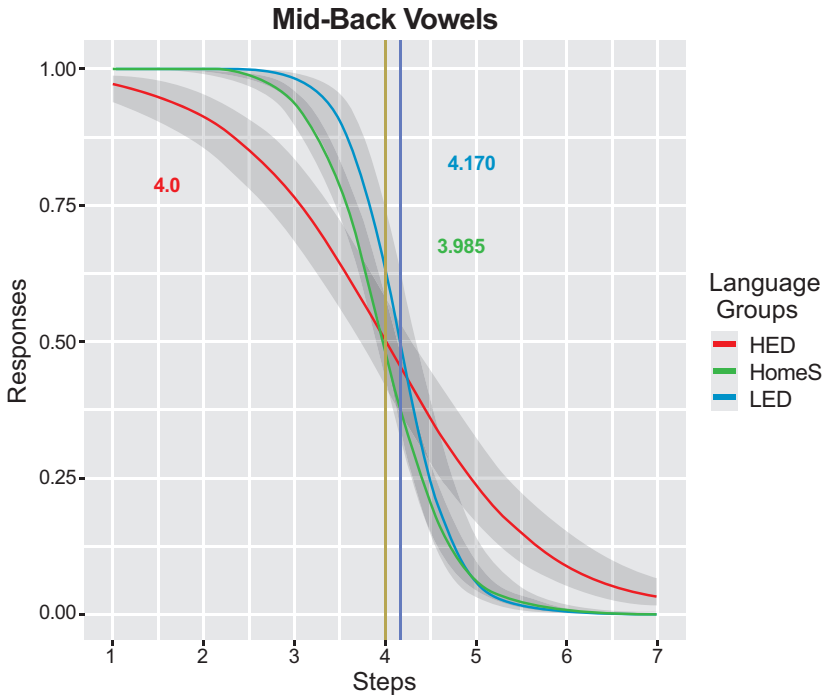


FIGURE 12 Curves representing the logistic regression model with group-based average slopes
*Note*. The vertical bars represent phonetic boundary crossover points. The HED group is represented in red, the LED group is represented in blue, and the HomeS group is represented in green. The prediction bands (95% confidence) indicated residual variance. On the y-axis, the number 1 corresponds to /ɔ/'s end point and 0 corresponds to /o/'s end point.

## 6 Discussion and Conclusions

Data analyses started with a series of correlations whose results showed, for instance, that Portuguese language skills were negatively correlated with language dominance and positively correlated with connection with language and culture (except for writing). However, the use of Portuguese by HSs (measured through HSs' self-reported average percentage of Portuguese use per week with friends, family, work/school, and talking to themselves) did not significantly correlate with Portuguese skills or with connection with the language. On the other hand, the use of Portuguese was inversely correlated to language dominance (i.e., those who use more Portuguese were less dominant

in English), which indicated that more dominance in English may reduce HSs' use of Portuguese.

One might expect that the use of Portuguese would correlate with other variables, such as Portuguese language skills (e.g., HSs who used more Portuguese in their lives would demonstrate higher levels of Portuguese skills; see Mora, Keidel, & Flege, 2015), but this was not the case. Heritage speakers reported that they used Portuguese most of the time with their families (51% of the time per week), whereas they did not use much Portuguese in other contexts (i.e., 15% and 9% of the time per week when at work/school or with friends, respectively). One possible explanation for the lack of correlation may be that HSs of BP in this study did not use much Portuguese outside their homes, apparently because they did not have many opportunities to do so, rather than because of personal choice. In the area where the data were collected (i.e., upstate New York), there were few Brazilians scattered around the city, an absence of Brazilian/Portuguese restaurants/shops, and no organized Brazilian/Portuguese community. To make confident conclusions about the role of use of the HL, more information from the HSs, such as answers to the following questions, would be necessary: Have HSs interacted with the Brazilian/Portuguese community where they live, assuming that there is one? Have HSs used Portuguese in public spaces when presented with the opportunity to do so (e.g., purchasing items in a Brazilian shop or meeting a Brazilian student on campus)? Future studies could explore these contexts more deeply, including the investigation of the extent to which the lack of opportunities to use heritage BP beyond the family environment could affect its development.

The main goal of this study was to investigate whether HSs of BP who were born and/or grew up in the United States perceived Portuguese mid-vowel contrasts (i.e., /ɛ/-/e/ and /ɔ/-/o/) categorically (RQ1). The study also aimed to investigate whether the degree of English dominance (as revealed by BLP scores) affected HSs' categorizations (RQ2). It was hypothesized that HSs and HomeS would demonstrate differences in categorization since these groups had different experiences with Portuguese that could affect their vowel perception (see Rao and Ronquest, 2015). It was also hypothesized that the perception of BP mid-vowel contrasts by HSs who were LED would be more robust (i.e., closer to HomeS) than that of HSs who were HED (Gertken, Amengual, and Birdsong, 2014). The findings in this study partially support these hypotheses since HED (but not LED) categorized the Portuguese /ɔ/-/o/ contrast less robustly, whereas no difference was found in the categorization of the Portuguese /ɛ/-/e/ contrast among language groups. The difference was more robust between HED and HomeS (i.e., differences at steps 2, 3, 5, 6, and 7) than

between HED and LED (i.e., a difference at step 3 only). Therefore, HSs of BP can categorize Portuguese mid vowels (RQ1); however, degree of English dominance affected the perception of the /ɔ/-/o/ contrast, but only for HED (RQ2).

These results indicate that the identification of the Portuguese /ɔ/-/o/ contrast (but not of the /ɛ/-/e/ contrast) appears to be more difficult for HED HSs. An interesting question is why this difficulty appears for the mid-back vowels, but not for mid-front vowels. First, it is important to point out that this type of asymmetry in perception between mid-front and mid-back vowels has been found in other studies investigating these contrasts. In Amengual (2015), no difference among Catalan-dominants and Spanish-dominants was found for the identification of the Catalan /ɛ/-/e/ contrast along a 7-step continuum; however, Spanish-dominants had more difficulty than Catalan-dominants in identifying /ɔ/ on the continuum. Asymmetrical results were also found in Feiden, Alves, and Finger (2014), where high school Spanish speakers from Argentina who had studied BP at school participated in a perception test in which they heard Portuguese words with target vowels and were asked to click on the vowel they heard. Error analyses verified that significantly more mistakes were made for /ɔ/ than for /o/ (i.e., more instances when /ɔ/ was perceived as /o/), whereas there was no difference in error rate between the perception of /ɛ/ and /e/.

Nevertheless, other studies have failed to show this type of asymmetry; for instance, Spanish-dominants have shown to be significantly worse than Galician-dominants in identifying both the /ɛ/-/e/ and the /ɔ/-/o/ vowel contrast in Galician (Amengual & Chamorro, 2015), and to have significantly higher error rates than Catalan-dominants in discriminating both the /ɛ/-/e/ and the /ɔ/-/o/ contrast (Amengual, 2016a). In a study involving the perception of Portuguese vowel contrasts by Russian speakers who had learned Portuguese as a foreign language (Smirnova Henriques et al., 2019), listeners showed a small overall difference between correct responses for the /ɛ/-/e/ contrast (71%) versus the /ɔ/-/o/ contrast (67%), but such a difference was non-significant.

We might assume that HSs of BP were mapping, for instance, Portuguese /ɔ/ onto English /ɔ/; however, "vowels transcribed with the same symbol do not necessarily have identical phonetic quality" (Disner, 1978, p. 21). Second Language Linguistic Perception (L2LP; see van Leussen & Escudero, 2015), a model on the development of speech perception, makes predictions on patterns of L2 categorization, taking into consideration similarities and differences between L1 and L2 sounds on acoustic-phonetic levels. Tuninetti, Mulak, and Escudero (2020) showed that differences in acoustic-phonetic relationships in vowel contrasts between Australian English and BP accurately predicted perceptual

difficulties in minimal pairs; BP /ɛ/-/e/ and /ɔ/-/o/ were correctly predicted to be more difficult to perceive by Australian English speakers (i.e., worse performance) than other contrasts, such as BP /i/-/o/. It would be interesting to see whether acoustic information from BP and American English vowels (e.g., F1 and F2 measurements) would correctly predict difficulties in the perception of vowel contrasts (see Elvin, Escudero, & Vasiliev, 2014; Tuninetti, Mulak, & Escudero, 2020).

This study has demonstrated that aspects of HL phonology may be affected by HSs' degree of language dominance; that is, when categorizing HL vowel contrasts, differences in categorizations may surface when considering the nuances of degrees of language dominance. Broadly speaking, we hope that the present study has also contributed to the overall understanding of the complexity of phonetic/phonological systems of HLs. More specifically, by focusing on BP, we hope to have contributed to the narrowing of the gap in phonology-based studies involving under-researched HLs.

## Acknowledgment

I would like to thank Dr. Joseph Casillas (Rutgers University), Dr. Mike Child (Brigham Young University) and the anonymous reviewers for their invaluable help and feedback. Special thanks to the heritage speakers who enthusiastically participated in this study. As one of them shared with me, "*Me define, o português*" ('Portuguese defines me').

## Notes on Contributor

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